# ROOT CAUSE ANALYSIS

*Why Are We Losing Customers?*

Telco Customer Churn Analysis Project

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## 1. Executive Summary

This document presents the root cause analysis of customer churn for a telecom company. Using a dataset of 7,043 customers analyzed through Python, MySQL, and Power BI, four primary root causes of churn were identified. Each finding is supported by data evidence, a root cause explanation, and actionable fix recommendations.

| Key Metric | Value | Benchmark | Status |
|---|---|---|---|
| Overall Churn Rate | 26.54% | 15-20% | Above Danger Zone |
| Total Customers Lost | 1,869 | - | High Impact |
| Avg Charges — Churned | $74.44/month | - | Premium Segment |
| Avg Charges — Active | $61.27/month | - | Lower Spend |
| Highest Risk Segment | New + Month-to-month | - | 47% churn rate |
| Lowest Risk Segment | Loyal (48-72m) | - | 8% churn rate |

## 2. Root Cause Analysis — 4 Key Findings

### Finding 1 — Contract Type is the #1 Churn Driver

| Contract Type | Total Customers | Churned | Churn Rate |
|---|---|---|---|
| Month-to-month | 3,875 | ~1,655 | ~42% |
| One year | 1,473 | ~166 | ~11% |
| Two year | 1,695 | ~48 | ~3% |

### Evidence:

- Month-to-month customers churn at 42% — 14x higher than two-year contract holders
- 55% of all customers are on month-to-month contracts — majority are at risk
- Two-year contract customers churn at only 3% — lowest of all groups

### Root Cause:

Month-to-month contracts have zero switching cost — customers can leave at any time with no financial penalty. This makes them highly vulnerable to competitor promotions and pricing offers. There is no retention mechanism built into the contract structure.

### Recommended Fix:

- Offer 20-25% discount to month-to-month customers who upgrade to annual contract
- Introduce a loyalty rewards program for long-term contract holders
- During onboarding, present annual plan as default — month-to-month as secondary option
- Send targeted upgrade campaigns to month-to-month customers at 3-month mark

### Expected Impact:

Shifting 20% of month-to-month customers to annual contracts could reduce overall churn rate from 26.54% to below 18% within 12 months.

## Finding 2 — Broken Onboarding (New Customer Churn)

| Tenure Group | Customers | Churned | Churn Rate |
|---|---|---|---|
| New (0-12 months) | ~2,175 | ~1,022 | ~47% |
| Growing (12-24 months) | ~1,100 | ~385 | ~35% |
| Established (24-48 months) | ~1,800 | ~360 | ~20% |

| | | | |
|---|---|---|---|
| Loyal (48-72 months) | ~1,968 | ~157 | ~8% |

### Evidence:

- Customers in the 0-12 month tenure group churn at ~47% — highest of all groups
- Nearly 1 in 2 new customers leave before completing their first year
- Churn drops significantly after 24 months — customers who stay become loyal

### Root Cause:

New customers do not experience enough value in the critical early stages. The expectations set during the sales process are not being met by the actual service. Without a structured onboarding program, customers feel abandoned after signing up and are easily attracted to competitor offers.

### Recommended Fix:

- Implement a structured 90-day onboarding program with weekly touchpoints
- Assign a dedicated onboarding specialist for the first 30 days
- Send Day 7, Day 30, and Day 90 check-in emails automatically
- Offer new customer exclusive benefits in first 3 months to build habit
- Track early engagement metrics and flag low-usage customers within 30 days

### Expected Impact:

A structured onboarding program targeting the 0-12 month segment could reduce new customer churn from 47% to below 30%, recovering approximately 370 customers per cycle.

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## Finding 3 — Fiber Optic Pricing vs Quality Gap

| Internet Service | Customers | Churned | Churn Rate |
|---|---|---|---|
| Fiber optic | ~3,096 | ~1,297 | ~42% |
| DSL | ~2,421 | ~459 | ~19% |
| No internet service | ~1,526 | ~113 | ~7% |

### Evidence:

- Fiber optic customers churn at 42% — highest of all internet service types
- Fiber optic is the most expensive service yet has the worst retention rate
- DSL customers churn at only 19% despite being a lower-tier service

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- Avg monthly charges for churned customers ($74.44) are higher than active ($61.27)

### Root Cause:

Premium-priced customers have higher expectations for service quality, speed, and reliability. When the actual experience does not justify the premium price — due to outages, slower-than-advertised speeds, or poor customer support — they feel deceived and switch to competitors. The price-to-value ratio is misaligned for fiber optic customers.

### Recommended Fix:

- Conduct service quality audit for fiber optic infrastructure immediately
- Introduce a Service Level Agreement (SLA) guarantee — if speeds drop, customers get credit
- Review fiber optic pricing vs top 3 competitors in each region quarterly
- Send proactive satisfaction surveys after every fiber optic service interaction
- Offer complimentary speed upgrade to fiber customers in their first 6 months

### Expected Impact:

Addressing fiber optic service quality and pricing could reduce fiber churn from 42% to below 28%, retaining approximately 430 high-value customers per cycle.

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## Finding 4 — Low Feature Engagement = High Churn Risk

| TechSupport | OnlineSecurity | Churn Rate | Risk Level |
|---|---|---|---|
| No | No | 48.96% | Critical |
| No | Yes | 21.30% | Medium |
| Yes | No | 22.33% | Medium |
| Yes | Yes | 9.01% | Low |

### Evidence:

- Customers with no TechSupport AND no OnlineSecurity churn at 48.96%
- Customers using both TechSupport and OnlineSecurity churn at only 9.01%
- Feature users churn at 5x lower rate than non-feature users
- Over 2,000 customers flagged as Feature Gap risk in the dataset

### Root Cause:

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Customers who do not use product features have low switching costs — they are not dependent on the product for anything critical. Low engagement means low perceived value, making it very easy to cancel. Customers who use security and support features have built a dependency on the product, creating a natural retention mechanism.

### Recommended Fix:

- Offer free 30-day trial of TechSupport + OnlineSecurity bundle to all non-users
- Add in-app notification highlighting security risks for customers without protection
- Send feature education email series over 4 weeks to low-engagement customers
- Train customer service team to mention and demonstrate features on every call
- Add feature adoption milestone rewards — e.g. discount after activating 2 features

### Expected Impact:

Increasing feature adoption by 25% in the Feature Gap segment could reduce their churn rate from 48.96% to below 30%, retaining approximately 500+ customers.

## 3. Root Cause Summary

| Rank | Root Cause | Churn Rate | Customers at Risk | Priority |
|---|---|---|---|---|
| #1 | Month-to-month contract — no commitment | 42% | 3,875 | URGENT |
| #2 | Broken onboarding — new customers lost early | 47% | 2,175 | URGENT |
| #3 | Fiber optic price vs quality gap | 42% | 3,096 | HIGH |
| #4 | Low feature engagement — easy to leave | 48.96% | 2,000+ | HIGH |

## 4. Expected Impact if All Fixes Implemented

| Metric | Current State | Target (12 months) | Improvement |
|---|---|---|---|
| Overall Churn Rate | 26.54% | Below 15% | -11.54% |
| New Customer Churn | ~47% | Below 30% | -17% |
| Fiber Optic Churn | ~42% | Below 28% | -14% |
| Feature Gap Churn | ~48.96% | Below 30% | -18.96% |
| Customers Recovered | 0 | ~560 (30%) | +560 retained |

## 5. Methodology & Tools

| Tool | Usage in This Analysis |
|---|---|
| Python (Pandas, Seaborn, Matplotlib) | Data cleaning, EDA, churn rate calculations, visualizations |
| MySQL Workbench | SQL queries for segmentation, cohort analysis, leading indicators |
| Power BI Desktop | 3-page interactive dashboard — Executive Summary, Deep Dive, Playbook |
| Dataset | Telco Customer Churn — Kaggle — 7,043 customers, 21 columns |